Ex. No.:3 11-09-2026	Intelligent Email Assistant	URK23CS1105

Aim:

To develop an Intelligent Email Assistant using Ollama that generates context-aware customer replies by analyzing previous email interactions.

Problem Statement:

Develop an AI-powered Email Assistant capable of understanding customer queries, retrieving similar historical conversations, generating professional responses, improving tone, summarizing conversations, and storing generated replies.

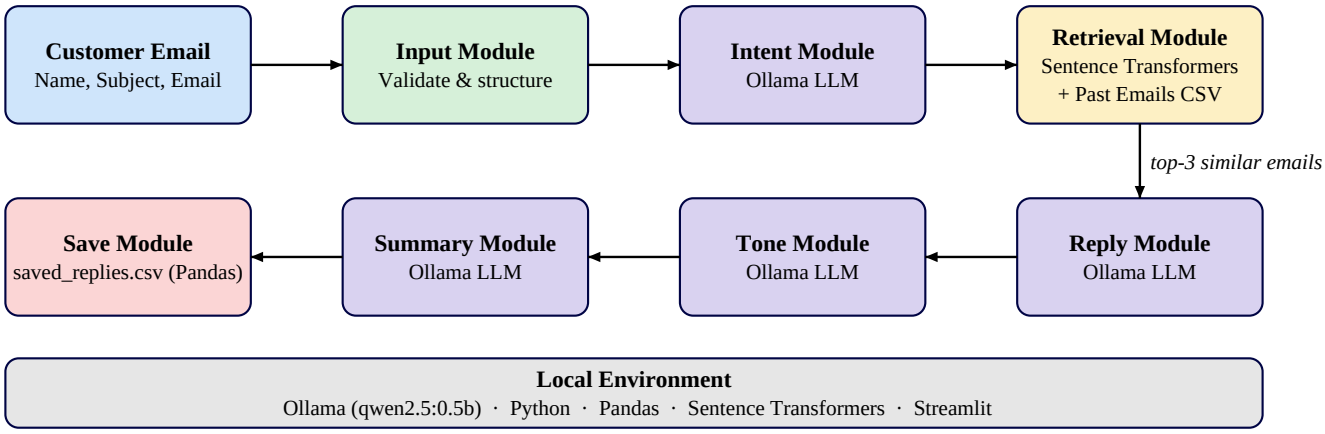
Software Requirements:

Software	Version
Python	3.14.2
Ollama	v0.34.0
VS Code	1.136.1
Pandas	3.0.5
Sentence Transformers	6.x

Input Parameters

Parameters	Description
Customer Name	Name of the customer who sent the email; used to address the reply.
Subject	Subject line of the customer email; used for intent detection and retrieval.
Email	Body of the customer email; analyzed to detect intent and find similar past emails.

System Architecture



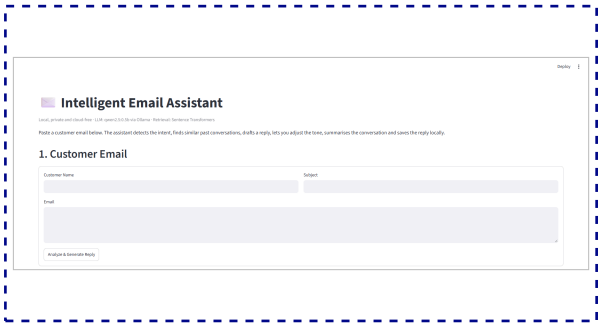
Module Design

Module	Responsibility
Input Module	Collects and validates the customer name, subject and email.
Intent Module	Uses Ollama to classify the email into a support category.
Retrieval Module	Finds the top-3 similar past emails using Sentence Transformers.
Reply Module	Drafts a professional reply using Ollama and similar past replies.
Tone Module	Rewrites the reply in a formal, friendly, empathetic or concise tone.
Summary Module	Summarizes the conversation with the customer.
Save Module	Stores the generated replies locally in a CSV file.

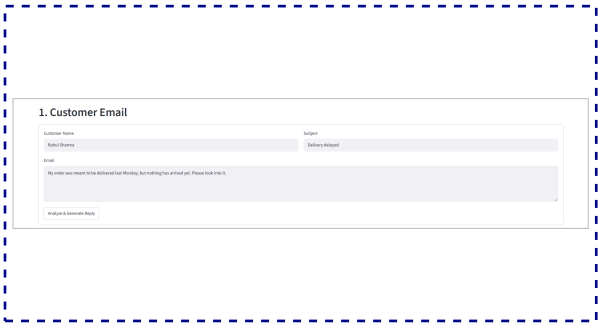
Implementation

GitHub Link: https://github.com/rishijayanath24-cpu/email-assistant-ollama
LinkedIn Post Link: https://www.linkedin.com/in/rishi-jayanath-a-346788321

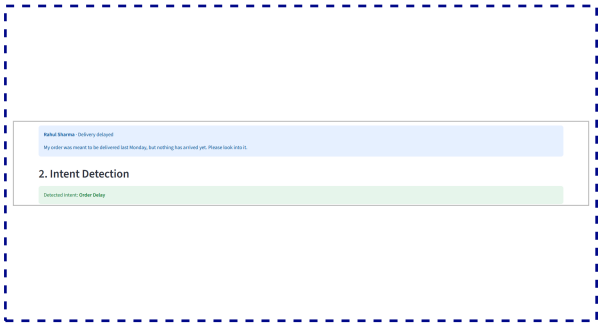
Output Screenshots:



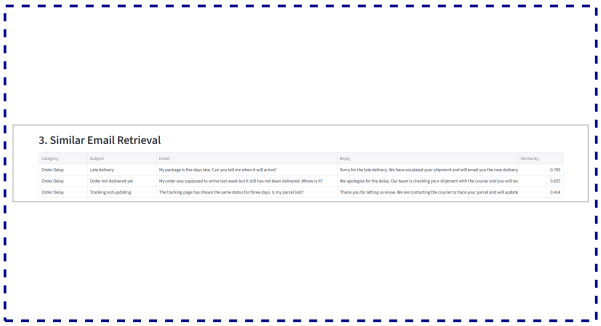
Home Screen



Customer Email



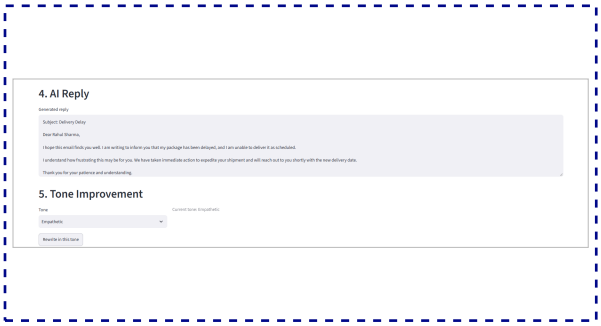
Intent Detection



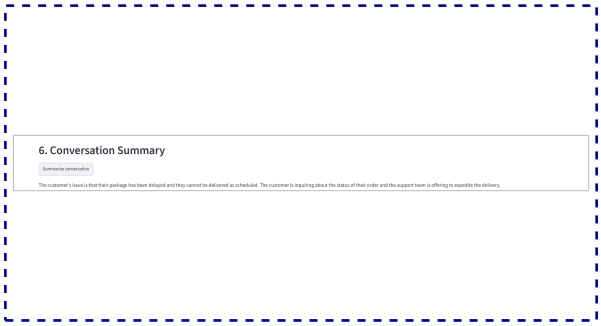
Similar Email Retrieval



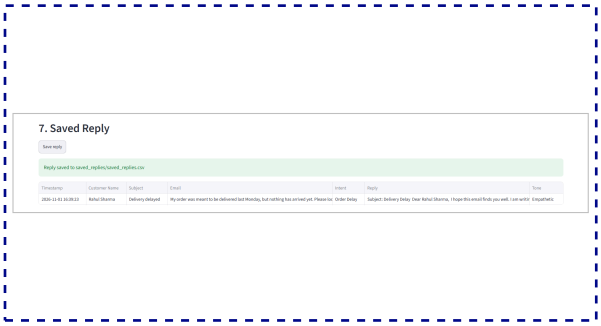
AI Reply



Tone Improvement



Conversation Summary



Saved Reply

Result:

Thus, an Intelligent Email Assistant was successfully developed using Ollama. The application understands customer emails, retrieves similar historical conversations, generates professional responses, rewrites replies in different tones, summarizes conversations, and saves generated replies locally without requiring cloud-based APIs.